

Dataset Construction Template (Draft v0.1)

A Lightweight Framework for Designing and Running Dataset Projects

This template is intended to guide the design and execution of dataset-building projects. It is deliberately lightweight and domain-agnostic so that domain experts and data scientists can adapt it to specific contexts.

Phase 1 — Define the Problem

1. Project Title

Name of dataset project.

2. Domain

Field or sector the dataset relates to.

Examples:

- horticulture
- energy systems
- transport
- environmental monitoring

3. Real-World Objective

What real-world problem should the dataset help address?

4. Intended AI Behaviour

What capability should a model trained on this dataset eventually have?

Example:

- classify objects
- detect events
- estimate conditions
- support decision-making

5. Domain Expert

Who defines what is meaningful or valid in this domain?

Name / affiliation.

6. Data Science Lead

Who is responsible for dataset design and quality?

Name / affiliation.

Gate: Phase 1 Sign-Off

Before proceeding to Phase 2, both the domain expert and data science lead should confirm: the real-world objective is achievable, the intended AI behaviour is realistic given the data type being considered, and there are no obvious blockers to feasibility. If either cannot sign off, revise Phase 1 before continuing.

Domain Expert sign-off: _____ Data Science Lead sign-off:

Phase 2 — Define Dataset Scope

7. Target Data Type

What type of data will be collected?

Examples:

- images
- audio
- sensor readings
- structured records

8. Core Data Categories

What are the main categories or labels expected in the dataset?

(To be defined with the domain expert.)

9. Scope Boundaries

Clearly define:

IN SCOPE

What should be included in the dataset.

OUT OF SCOPE

What should not be included.

Ambiguity here creates problems later.

Phase 3 — Design Data Capture

10. Data Collection Method

How will data be gathered?

Examples:

- volunteer capture
- automated sensors
- archival sources

11. Capture Guidelines

Basic rules for how data should be collected.

Defined by domain expert + data scientist.

12. Required Metadata

What contextual information should be recorded alongside each item?

Examples:

- time
- location
- device type
- environmental conditions

Phase 4 — Define Annotation

13. Label Definitions

Each label used in the dataset should be clearly defined.

Domain experts should provide examples where possible.

14. Uncertain Cases

Define how contributors should handle uncertainty.

Typical options:

- “Unknown”
- “Unclassifiable”
- “Other”

Important design decision: the domain expert and data scientist must explicitly distinguish between two types of uncertainty, and give contributors clear guidance on each. “The subject is ambiguous” (e.g. the item is genuinely hard to classify) is different from “the data is too poor to classify” (e.g. blurry image, missing signal). These may warrant separate

labels and will have different implications for downstream model training. Define both cases before annotation begins.

15. Annotation Roles

Who performs annotation?

Examples:

- volunteers
- trained annotators
- domain experts

Phase 5 — Quality Control

16. Review Process

How will annotations be checked?

Examples:

- peer review
- expert sampling
- automated checks

17. Calibration Items – Gold Set

Include a set of pre-labelled items, created by the domain expert, to test annotator accuracy. These should be embedded invisibly in the annotation queue so contributors cannot distinguish them from regular items. Calibration items are what separates a structured volunteer dataset from unverifiable noise — they should be treated as a core requirement, not an optional extra.

18. Disagreement Handling

Define in advance — before the pilot begins — what happens when annotators disagree. Two decisions must be made explicitly:

- At what threshold does disagreement trigger a process stop? For example: if annotators disagree on more than 20% of items, do you pause and revise the label definitions before continuing, or proceed and flag for expert arbitration?
- Who resolves disagreements that cannot be settled by peer review? Name a specific person or role. Do not leave this undefined.

These decisions should be recorded here before annotation begins.

Phase 6 — Pilot Execution

19. Micro-Pilot

Run a small version of the project before scaling.

Typical pilot size:

- small contributor group
- limited dataset size

Purpose:

identify problems in the process.

20. Process Review

After the pilot, review:

- capture protocol
- annotation clarity
- metadata completeness
- sources of bias

Diagnostic check: what percentage of items in the pilot were routed to “Unknown”, “Unclassifiable”, or “Other”? If the answer is close to zero, treat this as a warning sign. It may mean contributors are forcing ambiguous items into named categories rather than admitting genuine uncertainty. Labels that are too broad, or training that discourages uncertainty, both produce this pattern — and both degrade data quality. Review whether the label definitions are too permissive before scaling.

Phase 7 — Iteration

21. Identify Failure Points

Document where the process broke down.

Examples:

- unclear instructions
- ambiguous labels
- missing metadata
- systematic bias

22. Update Template

Revise the dataset design based on lessons learned.

Phase 8 — Scale

23. Full Dataset Execution

Run the dataset project with a larger group of contributors.

24. Dataset Assembly

Prepare the final dataset for use.

This may include:

- cleaning
- verification
- documentation

Phase 9 — Documentation

25. Dataset Description

Document:

- how the dataset was created
- who contributed
- known limitations

26. Intended Use

Define what the dataset is suitable for.

27. Known Limitations

All datasets have limitations.

These should be recorded explicitly.

Iterative Development Principle

This template is expected to evolve through repeated pilots.

Each dataset project should contribute improvements to the template so that future projects become easier and more reliable.

Intended Outcome

Over time, the goal is to develop a repeatable dataset construction process that can be embedded into structured workflows on the Volunteer Platform, enabling communities to produce high-quality datasets across multiple domains.